



American International University-Bangladesh (AIUB)
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PROJECT TITLE

Intelligent Pharmacy Inventory for Modern Pharmacies

A Software Engineering Project Submitted
By

Semester-->Summer-25-26		Section: E
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1. PROJECT PROPOSAL

1.1 Background to the Problem

The pharmacy industry plays a vital role in providing safe and timely access to medicines. However, many small and medium-sized pharmacies still rely on manual methods or basic software to manage their inventories. These traditional approaches often lead to inaccurate stock records, delayed updates, expired medicines remaining on shelves, and frequent stock shortages. As the number of medicines and customer demands continue to grow, manual inventory management becomes increasingly inefficient, time-consuming, and prone to human error.

One of the major challenges faced by pharmacies is maintaining the right balance between medicine availability and inventory cost. Overstocking increases storage expenses and causes financial losses when medicines expire, while understocking results in missed sales opportunities and inconvenience for customers. Pharmacists also spend significant time checking stock levels, updating sales records, tracking expiry dates, and preparing reports. These repetitive tasks reduce productivity and increase the possibility of mistakes.

Modern technologies such as artificial intelligence (AI), automation, and data analytics can significantly improve pharmacy inventory management. An intelligent inventory system can monitor medicine stock in real time, generate low-stock alerts, predict future demand based on sales history, identify slow-moving products, recommend suitable alternative medicines when a product is unavailable, and notify pharmacists before medicines expire. Such a system enables pharmacies to make better purchasing decisions, reduce waste, improve operational efficiency, and provide better customer service.

The proposed project, "**Intelligent Pharmacy Inventory for Modern Pharmacies**," aims to develop a web-based inventory management system that integrates intelligent features for efficient stock monitoring and decision-making. The system will allow pharmacy staff to manage medicines, inventory, purchases, sales, customer orders, expiry dates, and reports through a centralized platform. Customers will be able to search medicine availability and place orders, while the system will also provide automated notifications for low-stock and expiring medicines, helping pharmacy staff take timely actions.

Additionally, the system will generate analytical reports and inventory insights that assist pharmacy Admin in understanding sales trends, optimizing stock levels, and improving business performance. By replacing manual inventory processes with an intelligent digital solution, the proposed system will reduce human errors, improve inventory accuracy, save time, enhance customer satisfaction, and improve the overall management of pharmacy operations.

1.2 Target Group of Users

The proposed system is designed for the following users:

- **Admin:** To manage the overall system, monitor inventory, manage users, view sales performance, access analytical insights, and oversee pharmacy operations.
- **Pharmacists:** To manage medicine stock, check expiry dates, process sales, recommend alternative medicines when required, and ensure medicines are available for customers.
- **Inventory Managers:** To update stock information, record purchases, monitor inventory levels, manage medicine expiry dates, and maintain accurate inventory records.
- **Cashiers:** To process customer purchases, generate invoices, and automatically update inventory after each transaction.
- **Customers:** To search medicine availability, place orders, view order history, and receive notifications regarding order status and medicine availability.

The **Intelligent Pharmacy Inventory for Modern Pharmacies** will provide a secure, user-friendly, and efficient platform that improves inventory control, minimizes medicine wastage, enhances decision-making, and delivers better services and convenience to customers.